

MERGE_{NET}: DIFFERENTIABLE SPATIAL ROUTING WITH A PHYSICAL TOKEN BOTTLENECK

Anonymous authors

Paper under double-blind review

ABSTRACT

Modern visual recognition requires models to capture both fine-grained image detail and long-range context. Vision transformers meet these needs by representing images as patch tokens and modeling their interactions, but retaining every token across layers becomes costly at high resolution. Token merging reduces this cost by consolidating redundant tokens. Yet its success depends both on learning which tokens to combine without losing task-relevant information and on translating those decisions into a physically shorter sequence. DTEM learns the merging policy with dedicated merge embeddings and a continuous relaxation. However, its soft training path retains every token slot: it learns merge decisions without realizing physical compression, which still requires hard merging. MergeNet closes this gap with a local-to-latent architecture. Differentiable routing consolidates information according to the learned merge decisions; fixed-budget selection then physically shortens the sequence before global reasoning. To make this transition trainable, we combine differentiable budgeted selection with mass-aware attention over merged tokens. We evaluate MergeNet on ImageNet-1K image classification, using DeiT-S/8 as the dense baseline and comparing with ToMe and PiToMe at matched token budgets. At a $2\times$ carrier reduction, the trained MergeNet checkpoint obtains 81.360% top-1, compared with 82.246% for dense DeiT-S/8, 81.932% for post-training ToMe, and 81.742% for PiToMe; the matched DTEM-p8 adaptation reaches [VALUE]%. Among the 150-epoch ImageNet routing variants, radius-3 routing scores 1.670 percentage points above global routing and 0.598 points above flattened-window routing. A three-seed CIFAR-100 study also favors radius-3 routing over global, flattened-window, and degree-matched receiver controls. These results characterize the spatial routing variant and physical bottleneck. Under the common trained-checkpoint protocol, the accuracy-efficiency comparison is [comparison summary].

1 INTRODUCTION

Vision transformers use patch tokens to represent image detail and self-attention to integrate information across the image. Processing a fine patch grid through every block, however, makes this global reasoning expensive. Token merging reduces the sequence by combining patches rather than simply removing them (Bolya et al., 2023). A learned merger thus faces two linked requirements: learn where information should flow, and physically reduce the sequence seen by downstream blocks during training.

Differentiable merging does not automatically solve this execution problem. DTEM learns dedicated merge embeddings through continuous grouping and merging (Lee & Hong, 2024). Its relaxed operators retain all token slots; for end-to-end training, DTEM alternates soft updates of the merge embeddings with hard, token-reducing updates of the backbone. Thus DTEM uses both continuous policy learning and physical compression during training, but in separate update paths. We ask whether one forward pass can propagate gradients through soft routing while downstream blocks process an exact, physically shorter sequence.

We study this question with MergeNet, a local-to-latent vision transformer. Six local blocks encode the full patch grid, after which soft mass transport learns how patch information moves between slots. A hard gather then selects K carriers from N patches, so the subsequent latent blocks receive

a physically shorter tensor during both training and inference. A residual cross-attention read lets those carriers access the full-grid local features before global latent reasoning. The gather indices are discrete; a straight-through weight connects the selected features to continuous routing scores without claiming to differentiate the indices themselves.

The routing support follows the original two-dimensional patch grid. A donor can send mass only to eligible receivers within radius R , while learned similarities determine the weights and the resulting mass determines which carriers survive the gather. This constraint preserves spatial proximity without prescribing a fixed pooling assignment. We compare the resulting routing variant with global, flattened-window, and degree-matched alternatives.

Carrier mass also has a defined role after selection. Following proportional attention in ToMe (Bolya et al., 2023), MergeNet adds each carrier’s $\log m$ to its attention key logits. This per-key bias is rank one across queries; the released operator reproduces it by adding one query/key dimension and using the original softmax scale in an unmodified FlashAttention call (Dao et al., 2022). We use an existing threshold-based smooth top- k relaxation (Su, 2024) for the continuous selection scores. Our contribution is the way these components connect soft spatial transport to an exact-budget latent path, together with the mass-bias implementation.

The experiments characterize the spatial-routing variant, with a clear limit on the overall performance claim. In a single-seed, 150-epoch ImageNet-1K study, radius-3 routing scores 1.670 percentage points above the global variant and 0.598 points above the flattened-window variant. A three-seed CIFAR-100 study also favors radius 3 over global, flattened, and degree-matched routing. At a matched final budget of 392 patches, the trained 300-epoch MergeNet checkpoint reaches 81.360% ImageNet top-1, below post-training ToMe’s 81.932%; the uncompressed DeiT-S/8 reaches 82.246%. The matched DTEM-p8 adaptation reaches [VALUE]% top-1. Together, the routing and cross-method results establish the behavior of the physical bottleneck; the common-protocol accuracy–latency comparison is [comparison summary].

Our contributions are:

- a local-to-latent architecture that combines differentiable spatial mass transport with an exact-budget gather before latent processing;
- comparisons of original-grid routing with global, flattened-window, and degree-matched alternatives, including a three-seed CIFAR-100 study; and
- an exact query/key reparameterization for mass-aware attention with an unmodified FlashAttention call, accompanied by separate reporting of accuracy, token budget, arithmetic cost, and measured runtime.

2 RELATED WORK

Token reduction. DynamicViT learns token retention (Rao et al., 2021), while A-ViT assigns adaptive depth through halting (Yin et al., 2022). ToMe uses bipartite soft matching and can compress pretrained or jointly trained transformers (Bolya et al., 2023); PiToMe (Tran et al., 2024), MCTF (Lee et al., 2024), and SAD-TM (Xie et al., 2026) modify its matching signal or merge schedule. DiffRate treats layerwise budgets as another learned variable (Chen et al., 2023). Equal final counts therefore do not imply equal layerwise computation.

Differentiable merging. DTEM is the closest predecessor: it decouples the merge representation from backbone features and trains it through relaxed grouping (Lee & Hong, 2024). MergeNet builds on its OpenToMe lineage (Westlake-AI, 2025), but places continuous mass transport and a hard fixed-budget gather in one forward path. The soft top- k relaxation follows Su (2024); it is not claimed as a new selection operator. Proportional attention using token size is also established by ToMe. Our systems addition is the rank-one query/key reparameterization for retaining that bias in FlashAttention.

Spatial structure and recovery. DSM and ALGM combine local and global merging (Heo et al., 2024; Norouzi et al., 2024); GTP-ViT propagates tokens over spatial and semantic edges (Xu et al., 2024). Recent reducers constrain matching along rows/columns or after space-filling reordering (Gong & Lis, 2025; Li & Xu, 2025). MergeNet instead constrains eligible routes by Euclidean

radius on the original grid. LookupViT uses cross-attention between compressed and high-resolution streams (Koner et al., 2024); our recovery read is an architectural component rather than a claimed general recovery method. Since local merging can reduce arithmetic without improving throughput (Heo et al., 2024), we keep analytic cost and measured runtime distinct.

3 METHOD

3.1 OVERVIEW

MergeNet connects differentiable routing to a physically shorter latent sequence within one forward pass. Given N image patches and a class token, a local encoder produces full-grid features $H^{(1)}, \dots, H^{(L)}$. Starting from $X^{(0)} = H^{(L)}$ and unit patch masses, L soft routing steps redistribute information while retaining every patch slot. A hard gather then selects K patch carriers. The carriers read the pre-transport features through cross-attention, and a latent encoder followed by a class head produces the prediction. Figure 1 shows the project schematic; the evaluated classifier uses its encoder-to-latent path.

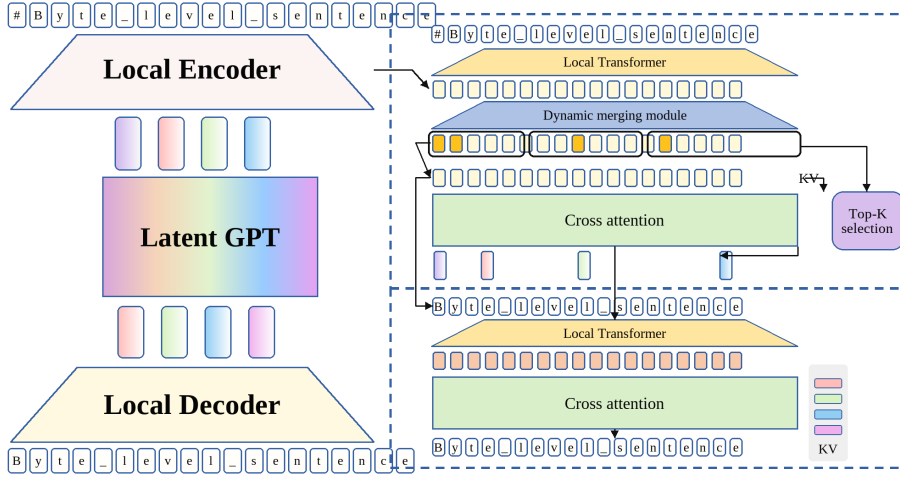


Figure 1: Project schematic of the local-to-latent design. The evaluated ImageNet classifier uses the encoder, dynamic merger, top- K selection, recovery cross-attention, latent encoder, and class head. The generic local decoder branch shown here is not used in this classifier.

3.2 SPATIAL MASS ROUTING

Spatial routing restricts where patch information may move without fixing the routing weights. Patch i keeps its coordinate u_i on the original two-dimensional grid. At routing step ℓ , patches are partitioned into disjoint donor and receiver sets A_ℓ and B_ℓ ; the class token participates in neither set. The eligible receivers for donor i are

$$\mathcal{E}_i^{(\ell)}(R) = \{j \in B_\ell : \|u_i - u_j\|_2 \leq R\}. \quad (1)$$

Coordinates remain tied to original slots as features move, so this rule constrains individual transport edges rather than the model’s complete receptive field.

The routing weights combine learned similarity with a soft donor decision. A learned projection of $H^{(\ell)}$ yields normalized metrics $z_i^{(\ell)}$. For eligible pairs, we compute

$$s_{ij}^{(\ell)} = \langle z_i^{(\ell)}, z_j^{(\ell)} \rangle / \tau, \quad q_{ij}^{(\ell)} = \frac{\exp s_{ij}^{(\ell)}}{\sum_{r \in \mathcal{E}_i^{(\ell)}(R)} \exp s_{ir}^{(\ell)}}. \quad (2)$$

The row energy $e_i^{(\ell)} = \sum_{j \in \mathcal{E}_i^{(\ell)}(R)} \exp s_{ij}^{(\ell)}$ enters a threshold-based smooth top- k operator (Su, 2024). Given the step budget k_ℓ , it produces a continuous donor weight $g_i^{(\ell)}$; the transport weight is $a_{ij}^{(\ell)} = g_i^{(\ell)} q_{ij}^{(\ell)}$ on eligible edges and zero elsewhere; a donor with no eligible receiver has zero assignment. The smooth weights provide a gradient path from the task loss to the routing metrics.

Soft transport updates mass and receiver features without removing the donor slot. With patch feature X_i and mass m_i , one step gives

$$\begin{aligned} m'_i &= m_i \left(1 - \sum_j a_{ij} \right), & X'_i &= X_i, \\ m'_j &= m_j + \sum_i a_{ij} m_i, & X'_j &= \frac{m_j X_j + \sum_i a_{ij} m_i X_i}{m'_j}. \end{aligned} \quad (3)$$

The sums cover eligible edges in the current donor–receiver partition. The implementation stabilizes residual fractions and denominators, while all patch slots remain present for the next routing step.

3.3 PHYSICAL BOTTLENECK AND RECOVERY

A hard gather turns the soft routing state into the shorter sequence used by later blocks. For compression factor λ , the patch-carrier budget is

$$K = N - \lfloor N(\lambda - 1)/\lambda \rfloor. \quad (4)$$

Let $\mathcal{I} = \text{TopK}(m_{\text{patch}}^{(L)}, K)$ contain the indices of the K largest patch masses, ordered before gathering. After normalizing the routed features, $\bar{X} = \text{Norm}(X^{(L)})$, we form the carrier sequence $C = [\bar{X}_{\text{cls}}; \bar{X}_{\mathcal{I}}]$. Thus C contains exactly $K + 1$ tokens, including the protected class token, in both training and inference.

The gather keeps its indices discrete while passing gradients to soft selection scores. A second smooth top- K operator applied to the final patch masses gives weights h_i . Each selected feature is multiplied by $1 + h_i - \text{stopgrad}(h_i)$, which equals one in the forward pass. This connects the selected features to the continuous scores without differentiating the index choice.

Recovery lets the carriers read information still available on the full grid. Using $H^{(L)}$ as the pre-transport key and value sequence, define

$$Z = C + \text{CrossAttn}(C, H^{(L)}, H^{(L)}). \quad (5)$$

The latent encoder processes Z , and its class-token output enters the classification head. Global latent self-attention therefore runs on $K + 1$ tokens, while recovery still reads the full-grid features.

3.4 MASS-AWARE LATENT ATTENTION

The selected carriers represent different amounts of transported mass. Following proportional attention in ToMe (Bolya et al., 2023), latent self-attention adds the log of each carrier mass to its key logit. For per-head query, key, and value matrices Q_a, K_a, V_a , head width d_h , and selected masses m_j , this is

$$\text{Attn}(Q_a, K_a, V_a; m) = \text{softmax} \left(\frac{Q_a K_a^\top}{\sqrt{d_h}} + \mathbf{1} \log(m)^\top \right) V_a. \quad (6)$$

The class token retains unit mass. This bias accounts for mass in attention; its contribution to classification accuracy requires a separate ablation.

The per-key bias can be encoded as one extra query/key dimension. Set $\hat{Q}_a = [Q_a, \sqrt{d_h} \mathbf{1}]$ and $\hat{K}_a = [K_a, \log m]$. With the original scale $1/\sqrt{d_h}$,

$$\frac{\hat{Q}_a \hat{K}_a^\top}{\sqrt{d_h}} = \frac{Q_a K_a^\top}{\sqrt{d_h}} + \mathbf{1} \log(m)^\top. \quad (7)$$

The released FlashAttention-compatible operator implements this identity after padding the head dimension. The reported ImageNet classifier uses the same bias through PyTorch’s scaled-dot-product attention; the operator identity alone does not establish a runtime gain.

4 EXPERIMENTS

We evaluate MergeNet on ImageNet-1K classification, then compare routing variants to study the spatial prior. We also examine how accuracy changes with the carrier budget and image resolution, and distinguish theoretical savings from measured inference time.

4.1 EXPERIMENTAL SETUP

Data and training. Our main experiments train on ImageNet-1K and report EMA top-1 accuracy on all 50,000 validation images. At 224×224 resolution, the DeiT-S/8 backbone (Touvron et al., 2021) starts from 784 patch tokens. Dense DeiT-S/8 and MergeNet train from scratch for 300 epochs with AdamW, an effective global batch of 1,024, cosine learning-rate decay, five warmup epochs, and the same augmentation family. The selected MergeNet run uses a learning rate of 7.5×10^{-4} ; dense DeiT uses 5×10^{-4} . The spatial-routing controls use the latter rate for 150 epochs. Appendix A gives the full recipes and run inventory.

Comparisons. MergeNet gathers 392 patch carriers after six full-grid local blocks. ToMe and PiToMe instead compress the same trained dense checkpoint at inference time. We set all three methods to 392 final patches, excluding CLS, while keeping their different compression schedules explicit. The patch-8 DTEM adaptation follows the common recipe (Table 1). For the resolution study, 384-pixel models fine-tune the 224-pixel EMA weights for 30 epochs, and 512-pixel models receive a further 15 epochs.

Evaluation. The compressed rows use full-validation evaluations of selected EMA checkpoints; the dense score is the best EMA training-summary endpoint. In particular, MergeNet scores 81.360% at 392 patches in the re-evaluation, compared with 81.368% for its best training-summary checkpoint. We keep these records distinct. The reported values are checked against training summaries and supplied evaluation records. The trained-checkpoint runtime comparison uses a common GPU, precision, and batch protocol, with each method’s accuracy-evaluation attention setting; existing synthetic benchmarks are reported separately in Section 4.4.

4.2 MAIN RESULTS

At the 392-patch operating point, MergeNet reaches 81.360% ImageNet top-1 accuracy (Table 1). ToMe and PiToMe, applied to the dense checkpoint after training, reach 81.932% and 81.742%; uncompressed DeiT-S/8 reaches 82.246%. Thus the proposed model learns its routing policy while processing a physically shorter latent sequence, although the available ToMe configuration retains more classification accuracy at the same final patch count. The matched DTEM-p8 adaptation reaches [VALUE]% top-1.

Table 1: ImageNet-1K top-1 accuracy at 224 pixels. Compressed rows target 392 final patches, excluding CLS.

Method	Checkpoint / path	Final patches	Top-1 (%)
DeiT-S/8	trained dense	784	82.246
DeiT-S/8 + ToMe	post-training	392	81.932
DeiT-S/8 + PiToMe	post-training	392	81.742
MergeNet ($R = 3$)	trained bottleneck	392	81.360
DTEM-p8	common-recipe adaptation	392	[VALUE]

The budget sweep shows how these models behave away from the default operating point (Figure 2). Using the same MergeNet checkpoint, increasing the final budget from 196 to 523 patches raises top-1 from 80.228% to 81.380%. ToMe remains ahead at each of the five 224-pixel budgets and at the five 384-pixel fine-tuned budgets shown in the figure. The complete sweep contains 20 matched final-budget settings, all with the same ordering (Appendix B). These are evaluations of fixed checkpoints; they describe accuracy as the inference budget changes rather than the outcome of retraining at each budget.

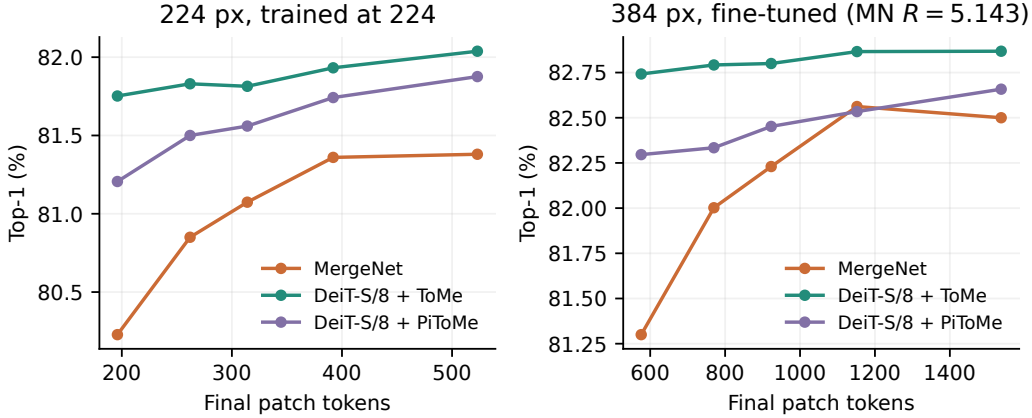


Figure 2: ImageNet top-1 across final patch budgets at 224 pixels (left) and after 384-pixel fine-tuning (right). Lines connect evaluations of fixed checkpoints.

Fine-tuning at higher resolution reduces the gap to dense DeiT at 384 pixels, but the gap grows slightly at 512 pixels (Table 2). The best EMA difference changes from -0.878 points at 224 pixels to -0.394 at 384 pixels and -0.450 at 512 pixels. The last MergeNet model descends from the radius-scaled 384-pixel branch, so this sequence describes the selected configurations rather than one unchanged-radius trajectory.

Table 2: Best EMA ImageNet-1K top-1 (%) across resolutions. Gap is MergeNet minus dense in percentage points.

Pixels	Schedule	DeiT-S/8 (%)	MergeNet (%)	Gap (pp)
224	300 scratch	82.246	81.368	-0.878
384	300+30	82.976	82.582	-0.394
512	300+30+15	83.068	82.618	-0.450

4.3 ABLATION STUDIES

Among the tested MergeNet routing variants, the original-grid radius-3 configuration scores above the global and flattened-window configurations. With the same 150-epoch ImageNet recipe and learning rate, radius 3 reaches 78.888% top-1, compared with 77.218% for global routing and 78.290% for a flattened width-8 window (Table 3). Radius 2 is slightly higher at 78.942%; these single-seed results identify a useful routing configuration, not an optimal radius. The global arm retains MergeNet’s architecture and is an internal variant rather than a DTEM reproduction. The variants also use their respective scoring implementations, detailed in Appendix A.

Table 3: ImageNet routing variants at 224 pixels. All rows use the same 150-epoch recipe and learning rate 5×10^{-4} .

Routing / probe	LR (10^{-4})	Best (%)	Final (%)
Global	5.0	77.218	77.216
Flat window 8	5.0	78.290	78.278
Spatial $R = 2$	5.0	78.942	78.942
Spatial $R = 3$	5.0	78.888	78.888

Radius also changes the number of eligible receivers. We therefore add a CIFAR-100 control that permutes radius-3 receivers while preserving donor candidate counts and receiver degrees (Appendix A).

Across three seeds on resized CIFAR-100, radius 3 has the highest epoch-199 EMA top-1 mean among the four routing policies (Table 4). Its paired gain over the degree-matched control is 1.42 ± 0.30 points, with all three seed-wise differences positive. This supports the spatial prior within the tested setting; the replicated result does not estimate ImageNet seed variance.

Table 4: CIFAR-100 epoch-199 EMA top-1 over seeds 42–44. Δ is the paired $R = 3$ minus control difference (pp).

Routing	Top-1 (%)	$R = 3$ minus control (pp)
Global	68.48 ± 0.10	$+2.11 \pm 0.87$
Flat window 8	69.29 ± 0.54	$+1.30 \pm 0.47$
Degree-matched	69.17 ± 0.62	$+1.42 \pm 0.30$
Spatial $R = 3$	70.59 ± 0.88	—

The component study tests whether the trained classifier depends on mass-aware attention and full-grid recovery. With the same checkpoint, carrier selection, and 50,000 validation images, disabling the $\log m$ key bias changes top-1 by [VALUE] percentage points; bypassing residual recovery changes it by [VALUE] points, and disabling both changes it by [VALUE] points (Appendix E). These inference-only interventions measure dependence of the trained model, rather than the effect of training without each component.

4.4 EFFICIENCY

The routing gains must be considered alongside the work required to produce the shorter sequence. MergeNet reduces the latent-stage input from 784 to 392 patches, but its six local blocks still process the full grid and recovery reads those features again. Accounting for these operations gives 14.13 GMAC per image at 224 pixels, compared with 22.41 GMAC for dense DeiT-S/8, a 36.9% reduction. The estimate excludes indexing and data movement, so it does not predict inference latency by itself.

Existing synthetic benchmarks show why the measured implementation matters. On an H20-3e at batch 64, random-weight MergeNet takes 45.71 ms versus 42.68 ms for dense DeiT-S/8 at 224 pixels, while the 384-pixel comparison is 136.54 versus 170.61 ms. An independent A100 benchmark does not reproduce the latter ordering on its software stack (Appendices C and D). These are architecture-only measurements with random weights. The 384-pixel H20 ToMe timing also disables proportional attention, unlike its accuracy evaluation. They therefore cannot be paired with checkpoint accuracy. Under the common-protocol trained-checkpoint evaluation, MergeNet records [VALUE] ms batch-1 latency, [VALUE] images/s batch-64 throughput, and [VALUE] GiB peak allocated memory. The comparison uses the selected checkpoints under a common GPU and precision, with each method’s accuracy-evaluation attention setting and achieved token counts recorded (Appendix E).

4.5 VISUALIZATION

The spatial prior also makes routing behavior inspectable. Figure 3 shows donor-to-receiver edges and final carrier locations for fixed ImageNet examples, while Figure 4 summarizes selection frequency and routing distance over 1,000 images. The passive-trace check gives **[logit, carrier-index, and mass parity]** (Appendix F).

5 DISCUSSION

The results separate the routing mechanism from overall model accuracy. The exact gather shortens the latent sequence, and the original-grid variant scores above the tested global and flattened-window variants on ImageNet and three-seed CIFAR-100. At equal final patch counts, the complete classifier trails ToMe on ImageNet. The routing-variant and cross-method comparisons therefore answer different questions.

Deployment cost and component attribution require separate measurements from the carrier count. Full-grid processing and routing overhead affect runtime; the trained-checkpoint comparison gives

[latency and throughput interpretation]. ImageNet geometry uses one seed, while the replicated locality test uses resized CIFAR-100. The fixed-checkpoint component tests give [mass-bias and recovery interpretation]. The routing visualization describes transport without implying semantic boundary preservation.

6 CONCLUSION

MergeNet combines differentiable spatial transport with an exact-budget gather so that learned routing feeds a shorter latent sequence. The radius-3 variant scores above the tested global and flattened variants; a three-seed comparison also favors it over degree-matched routing. The current ImageNet result is 81.360% top-1 at 392 final patches, with a clear accuracy gap to ToMe at that budget. Under a common inference protocol, the trained-checkpoint accuracy–efficiency tradeoff is [summary relative to baselines]. Reducing full-grid and routing overhead remains a direction for further work.

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A IMPLEMENTATION DETAILS

Table 5 gives the training settings for the main ImageNet experiments. All runs use an effective global batch of 1,024. The 150-epoch routing variants share a learning rate of 5×10^{-4} ; the selected 300-epoch MergeNet run uses 7.5×10^{-4} . EMA top-1 is measured on the 50,000-image validation set.

Table 5: Training settings for the ImageNet-1K experiments.

Setting	224-pixel scratch	384/512-pixel fine-tune
Epochs	150 or 300	30 at 384; 15 at 512
Effective global batch	1024	1024
Optimizer	AdamW	AdamW
Learning rate	5×10^{-4} or 7.5×10^{-4}	10^{-5}
Minimum learning rate	10^{-5}	10^{-6}
Weight decay	0.05	10^{-8}
Warmup epochs	5	1
Warmup learning rate	10^{-6}	10^{-6}
EMA decay	0.99996	0.99996
Mixup / CutMix	0.8 / 1.0	0 / 0
Label smoothing	0.1	0.1
Random erasing	0.25	0.25
RandAugment	rand-m9-mstd0.5-inc1	same
Gradient clip norm	1.0	1.0
Validation crop percentage	0.9	1.0
MergeNet auxiliary-logit weight	ramp to 0.05	0.05
Teacher / feature / routing distillation	disabled	disabled

Classifier. The patch-8 model has width 384, six heads, six local blocks, and six latent blocks. At 224 pixels, the local encoder processes 784 patch slots. Local attention spans 16 sequence positions on either side of a patch query; the CLS query attends globally. Spatial routing uses radius 3, a $384 \rightarrow 64$ metric projection, similarity temperature 1, and metric input-gradient scale 0.1. At compression factor 2, the six soft donor budgets are (65, 65, 65, 65, 65, 67); the hard gather retains 392 patches. The latent encoder does not merge further. Stochastic depth is 0.1, with separate local and latent ramps.

Routing and selection. Training partitions donors and receivers randomly for each image; evaluation alternates deterministic odd/even partitions across routing steps. The threshold-based soft top- k uses scale 30 and normalizes energies by their max-minus-min range. The final gather indices are detached, and selected carriers are ordered by a detached, mass-weighted center of their original flattened indices. Spatial scoring uses sparse neighbor lists; the global and flattened-window controls use their respective historical dense-scoring paths. At training time, an auxiliary full-grid readout contributes up to 0.05 to the latent CLS logits. Its weight begins ramping at epoch 10 for the 150-epoch runs and epoch 20 for the 300-epoch run, reaching 0.05 over 10 and 20 epochs, respectively. It is disabled at evaluation.

CIFAR-100 controls. The three-seed routing comparison uses 12 independent 200-epoch runs: global, flattened-window, radius-3, and degree-matched routing for seeds 42–44. Native 32×32 images are resized to 224 pixels. The primary endpoint is epoch-199 EMA top-1 on the 10,000-image test set. Each training and evaluation batch contains 200 images. The degree-matched arm permutes receiver columns of the realized radius-3 support using fixed receiver priorities; it preserves each donor’s candidate count and the receiver-degree multiset. Table 4 reports means and sample standard deviations across seeds, with paired same-seed differences.

B ACCURACY BUDGETS

Table 6 gives the values behind Figure 2. The MergeNet checkpoint uses radius 3 at 224 pixels and radius 5.143 after 384-pixel fine-tuning. Each row matches final patch counts, excluding CLS, but

not the layerwise sequence lengths: ToMe and PiToMe merge across blocks, whereas MergeNet gathers once between stages. All points evaluate fixed checkpoints at different inference budgets.

Table 6: ImageNet top-1 (%) at matched final patch counts. Δ is ToMe minus MergeNet in percentage points.

Checkpoint / pixels	Patches	MergeNet	ToMe	PiToMe	Δ ToMe
224 trained / 224	523	81.380	82.038	81.876	+0.658
224 trained / 224	392	81.360	81.932	81.742	+0.572
224 trained / 224	314	81.074	81.814	81.560	+0.740
224 trained / 224	262	80.850	81.830	81.500	+0.980
224 trained / 224	196	80.228	81.752	81.206	+1.524
384 tuned / 384	1537	82.500	82.868	82.658	+0.368
384 tuned / 384	1152	82.562	82.866	82.534	+0.304
384 tuned / 384	923	82.230	82.800	82.452	+0.570
384 tuned / 384	770	82.002	82.792	82.334	+0.790
384 tuned / 384	576	81.300	82.742	82.296	+1.442

C SYNTHETIC TIMING

The measurements in this section use randomly initialized models to isolate architecture and implementation cost. They do not use the trained checkpoints that produce the accuracy results in Tables 1 and 6. On an H20-3e at batch 64 and 224 pixels, MergeNet takes 45.71 ms per forward pass, versus 42.68 ms for dense DeiT-S/8 (Table 7).

Table 7: H20-3e synthetic timing at 224 pixels, batch 64, with random weights and fp16 autocast. Train time includes forward, loss, backward, and AdamW update; inference is forward only. Memory is peak allocated GiB.

Model	Params (M)	Train (ms)	Infer (ms)	IQR (ms)	Train GiB	Infer GiB
DeiT-S/16	22.05	37.62	9.99	0.005	2.472	0.367
DeiT-S/8	22.06	149.56	42.68	0.099	8.555	0.799
MergeNet, $R = 3$	23.10	154.87	45.71	0.073	10.071	1.125

At 384 pixels, a separate radius-3 H20-3e run measures 136.54 ms for MergeNet and 170.61 ms for dense DeiT-S/8 at batch 64. The radius-5.143 comparison with ToMe in Table 8 uses a different setup: ToMe proportional attention is disabled, achieved budgets differ slightly, and token-counting hooks remain active. At the approximately half-size budget, ToMe processes 1,151 patches in 129.81 ms and MergeNet processes 1,152 in 155.25 ms. These measurements cannot be paired with checkpoint accuracy to form an accuracy–latency comparison. The common trained-checkpoint benchmark is specified in Appendix E.

Table 8: H20-3e synthetic inference at 384 pixels, batch 64, fp16 autocast, random weights, and MergeNet radius 5.143. ToMe proportional attention is disabled and token-counting hooks remain active. Memory is peak reserved GiB; patch counts exclude CLS.

ToMe patches	MN patches	ToMe ms	MN ms	ToMe GiB	MN GiB
1536	1538	147.30	178.17	2.793	3.619
1151	1152	129.81	155.25	2.764	3.564
922	923	122.44	143.02	2.746	3.721
769	770	118.17	136.59	2.738	3.719
575	576	106.45	129.38	2.723	3.557

D A100 TIMING

An independent A100-SXM4-80GB experiment compares dense DeiT-S/8, ToMe, and the sparse-backend MergeNet with randomly initialized weights. All cells use batch 64, fp16 autocast, and 392 final patches at 224 pixels or 1,152 at 384 pixels for compressed methods. The measured forward pass excludes data loading, training, and diagnostic hooks. ToMe’s proportional-attention setting is shown explicitly; its enabled setting matches the accuracy evaluation. The BNHD rows use a local-attention tensor-layout change, rather than a different trained checkpoint.

Each cell uses ten warmups and thirty CUDA-event samples in each of three rounds. Table 9 reports the median of the three round medians.

Table 9: Synthetic A100 inference time (ms), batch 64. All weights are randomly initialized. The 384-pixel radius-5.143 model uses the same final patch budget as radius 3.

Model	224 px (ms)	384 px (ms)
Dense DeiT-S/8	30.09	109.43
ToMe, proportional attention on	34.42	133.36
ToMe, proportional attention off	29.71	96.05
MergeNet $R = 3$	49.16	144.97
MergeNet $R = 3$, BNHD	46.56	138.11
MergeNet $R = 5.143$	–	218.25
MergeNet $R = 5.143$, BNHD	–	211.40

On this hardware and software stack, the original MergeNet is slower than dense DeiT at both resolutions. BNHD reduces full-model radius-3 time by about 5%. ToMe’s proportional-attention setting also changes latency substantially. The A100 ordering differs from the 384-pixel H20-3e radius-3 result; the two experiments differ in hardware and software, so neither establishes an accelerator-independent speed ranking. These random-weight times are kept separate from the trained-checkpoint accuracy and common-protocol benchmark.

E CHECKPOINT EVALUATION

The common inference benchmark uses the same GPU, precision, and measurement procedure for all methods. Each method retains its accuracy-evaluation attention setting, and achieved token counts are recorded. The benchmark reports full-validation accuracy separately from batch-1 latency, batch-64 throughput, and peak allocated memory. Table 10 gives the trained-checkpoint measurements; the synthetic random-weight timings in Appendices C and D are reported separately.

Table 10: Common-protocol trained-checkpoint inference at 224 pixels. B1/B64 denote batch sizes 1 and 64.

Method	Patches	B1 (ms)	B64 (im/s)	B64 alloc. (GiB)
Dense DeiT-S/8	784	[VALUE]	[VALUE]	[VALUE]
DTEM-p8	392	[VALUE]	[VALUE]	[VALUE]
DeiT-S/8 + ToMe	392	[VALUE]	[VALUE]	[VALUE]
DeiT-S/8 + PiToMe	392	[VALUE]	[VALUE]	[VALUE]
MergeNet ($R = 3$)	392	[VALUE]	[VALUE]	[VALUE]

The component test uses the same trained MergeNet EMA checkpoint for all four rows of Table 11. The mass-bias intervention sets only the latent log-mass key bias to zero; the recovery intervention bypasses only residual cross-attention. The native score is checked against the common inference evaluation. Paired intervals compare per-image correctness on the same 50,000 validation examples and do not represent training-seed uncertainty.

Table 11: Inference-only component test at 224 pixels. Δ and its paired 95% interval are relative to native top-1.

Inference setting	Top-1 (%)	Δ top-1 (pp), 95% CI
Native (mass + recovery)	[VALUE]	—
Mass bias off	[VALUE]	[VALUE]
Recovery off	[VALUE]	[VALUE]
Both off	[VALUE]	[VALUE]

F ROUTING VISUALIZATION

The visual examples are selected before inspecting the router. We choose one ImageNet validation image per class using a fixed SHA-256 ordering of image paths, giving 1,000 images in total. From this set, the main figure shows the first correctly classified and the first misclassified image under the same ordering. Passive tracing is checked against untraced predictions, top- K indices, and token mass; the check gives [parity summary]. If no error occurs in the selected set, the selection rule uses two correct cases.

The main figure pairs actual step-3 donor-to-receiver transport edges with the final 392 carrier locations for each selected image. The population figure summarizes per-image routing distances over all six steps and the selection frequency of each original 28×28 patch location across the 1,000 images. Carrier markers denote selected slots, not semantic regions. Mean route distance is [VALUE] patches, and the selection-frequency pattern is [population interpretation].

[Figure: E6: actual routing edges and final carriers for fixed examples]

Figure 3: ImageNet routing examples: step-3 transport edges and final carrier locations from the same trained checkpoint.

[Figure: E6: routing distances and carrier selection frequency]

Figure 4: Population view for 1,000 fixed validation images. Distances are measured in original-grid patches; selection frequency uses the 392 final carrier slots.